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# MAKERERE UNIVERSITY BUSINESS SCHOOL

## FINAL EXAMINATION FOR THE DEGREE OF BACHELOR OF BUSINESS COMPUTING OF MAKERERE UNIVERSITY ACADEMIC YEAR 2016/2017

**COURSE NAME: SYSTEMS ANALYSIS AND DESIGN. YEAR OF STUDY: 2**

**COURSE CODE: BUC 2105 DATE: 27/11/2016**

**SEMESTER: 1. TIME: 2:00 PM – 5:00 PM**

Answer any **FOUR** questions

**Question 1**

a) The system development lifecycle (SDLC) is the process of understanding how an information system can support business needs, designing the system, building it and delivering it to the users. Discuss the four fundamental phases and sub-phases that are in the SDLC. **(10 marks)**

b) There are different models or methods that can be used in the development of information systems. With illustrations discuss each of the following system development methods. marks each)

i) Waterfall model II) V-model iii) Time boxing processing model

**Question 2**

## Question 2

a) The conceptual model, logical data model, and the physical data model have specific features that define them. Select the appropriate feature for each model. ( the correct selection will have +0.5 mark and the wrong selected feature will have a penalty of -0.5 mark) (5 marks)

| Feature | Physical data model | Conceptual model | Logical data model |
|---|---|---|---|
| Entity name | | | |
| Entity relationship | | | |
| Attribute | | | |
| Foreign key | | | |
| Validation rule | | | |
| Field size | | | |
| Null value | | | |





b) Many organizations have invested in computer networks. Discuss the advan disadvantages of a computer network in the management of information in an organization. (10 marks)

c) As a system analyst in your organization, you have been requested by your supervi design a network for your organization. Discuss the various steps you will undertake have a design that is acceptable by your supervisor. (10 marks)

## Question 3

## Question 3

a) Using an illustration, explain the role played by the primary key and foreign key in the development of a relational database. (5 marks)

b) Security of information systems in organisations is very critical. Discuss the three categories of security controls that can be implemented in an organization. (10 marks)

c) Given 3 entities of Employer, Department and Staff in an organisation. With illustrations, explain the conceptual data model, logical data model and physical data model using these entities in an organisation. (10 marks)

## Question 4

Human parts play a significant role in input design. Inputs should be as simple as possible and designed to reduce the possibility of incorrect data being entered. The needs of system users must be considered. With this in mind several human factors should be evaluated. The volume of data to be input should be minimized. The more data that is input, the greater the potential number of input errors and the longer it takes to input that data.

a) Explain any five general principles that should be followed when designing forms for data input (10 marks)

b) Explain human engineering factors that should be incorporated in the input design (10 marks)

c) Give any five output design guidelines (5 marks)

## Question 5

a) Describe three levels of software requirements (9 marks)

b) Good questionnaires are designed. If you write your questionnaires without designing them first you reduce your chances of success. Explain all the steps involved in design of good questionnaires (6 marks)

c) Explain how PIECES framework can be used for software requirements elicitation (10 marks)

Question 6

a. Briefly describe a complete computer-based information system in terms of its components (5 marks)

b. Justify with five reasons the importance of studying systems analysis and design (5 marks)

c. What roles do system users and system owners play in information systems development project? (4 marks)

d. Explain the term "Problem" in three different perspectives in information systems development (6 marks)

e. Why must an organization consider information systems maintenance as an important activity (5 marks)

THE END